DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-D32: The Fitness centre Case [20 marks]

Briefing: You are the data detective hired by the owner of a fitness centre. The file holds 15 daily records from the training season, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
outdoor_temp	Outdoor temperature (deg C)	injuries	Injury reports
gym_visits	Gym visits	trainers	Trainers on duty
shakes_sold	Protein shakes sold	towel_loads	Towel laundry loads
load_dev	Training load minus optimum (units)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(94142)
n <- 15
outdoor_temp <- round(rnorm(n, 28.6, 5.3), 1)
gym_visits <- round(209 + 52.7 * (0.93 * (outdoor_temp - 28.6) / 5.3 + 0.37 * rnorm(n)), 0)
shakes_sold <- round(57 + 15.8 * (0.84 * (outdoor_temp - 28.6) / 5.3 + 0.54 * rnorm(n)), 0)
load_dev <- round(seq(-17.6, 17.6, length.out = n)[order(runif(n))], 1)
injuries <- round(1.9 + 8.8 * (load_dev / 17.6)^2 + rnorm(n, 0, 0.3), 0)
trainers <- round(rnorm(n, 6, 1.6), 0)
towel_loads <- round(17 + 5.3 * (0.13 * (trainers - 6) / 1.6 + 0.99 * rnorm(n)), 0)
trainers[11] <- 11; towel_loads[11] <- 33 # row 11 was typed in by a trainee
case <- data.frame(outdoor_temp, gym_visits, shakes_sold, load_dev, injuries, trainers, towel_loads)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot injuries against load_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot towel_loads against trainers. Find the row that looks wrong and give its row number. Report r for all rows and

r without that row, and say what the single record did to r. [4]

5. The prime suspect. gym_visits and shakes_sold are strongly linked. Report r(gym_visits, shakes_sold), then r(outdoor_temp, gym_visits) and r(outdoor_temp, shakes_sold). Compute the partial correlation of gym_visits and shakes_sold controlling for outdoor_temp: correlate resid(lm(gym_visits ~ outdoor_temp)) with resid(lm(shakes_sold ~ outdoor_temp)). Write a two-sentence verdict: does gym_visits cause shakes_sold? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

Task 1: The matrix

n = 15 records, 7 variables, so there are 7 × 6 / 2 = **21 distinct pairs**.

	temp	**icecream**	**rescues**	**pool_dev**	**complaints**	**lifeguards**	**sunscreen**
temp	1.00	0.92	0.87	0.18	0.20	0.29	0.13
icecream	0.92	1.00	0.82	0.01	0.26	0.37	0.33
rescues	0.87	0.82	1.00	0.39	0.23	0.28	-0.09
pool_dev	0.18	0.01	0.39	1.00	-0.03	-0.21	-0.40
complaints	0.20	0.26	0.23	-0.03	1.00	0.21	0.15
lifeguards	0.29	0.37	0.28	-0.21	0.21	1.00	0.60
sunscreen	0.13	0.33	-0.09	-0.40	0.15	0.60	1.00

r

```r
set.seed(125879)
n <- 15
temp <- round(rnorm(n, 30.0, 4.3), 1)
icecream <- round(161 + 42.5 * (0.91 * (temp - 30.0) / 4.3 + 0.41 * rnorm(n)), 0)
rescues <- round(13 + 4.3 * (0.92 * (temp - 30.0) / 4.3 + 0.39 * rnorm(n)), 0)
pool_dev <- round(seq(-4.8, 4.8, length.out = n)[order(runif(n))], 1)
complaints <- round(2.1 + 12.9 * (pool_dev / 4.8)^2 + rnorm(n, 0, 0.9), 0)
lifeguards <- round(rnorm(n, 9, 2.1), 0)
sunscreen <- round(96 + 26.6 * (0.22 * (lifeguards - 9) / 2.1 + 0.98 * rnorm(n)), 0)
lifeguards[13] <- 16; sunscreen[13] <- 179
case <- data.frame(temp, icecream, rescues, pool_dev, complaints, lifeguards, sunscreen)

library(ggplot2); library(reshape2)
cm <- round(cor(case), 2); print(cm)
nrow(case); choose(7, 2)

ggplot(melt(cm), aes(Var1, Var2, fill = value)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", value)), size = 3.5) +
  scale_fill_gradient2(low = "#b2182b", mid = "white", high = "#2166ac",
                       limits = c(-1, 1), name = "r") +
  labs(title = "Correlogram: Coastal town case (n = 15)", x = NULL, y = NULL) +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Task 2: The strongest clues

- **Strongest positive:** temp and icecream, **r = 0.92**, which is **strong** (above 0.7).
- **Strongest negative:** pool_dev and sunscreen, **r = -0.40**, which is **moderate** (0.3 to 0.7).

Task 3: The deceptive zero

r

```
ggplot(case, aes(pool_dev, complaints)) +
  geom_point(size = 2.5) +
  geom_smooth(method = "lm", se = FALSE, colour = "grey40", linetype = "dashed") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), se = FALSE, colour = "red") +
  labs(title = "Complaints vs pool temperature deviation",
       x = "Pool temp minus 27 °C", y = "Swimmer complaints") +
  theme_minimal()

cor(case$pool_dev, case$complaints)
summary(lm(complaints ~ pool_dev + I(pool_dev^2), data = case))$r.squared
```

- **r = -0.03**, essentially zero.
- **Quadratic $R^2 \approx 0.976$**, so the U-shaped curve explains about 98% of the variation.

Why r misses it: r only measures straight-line association, and the left and right arms of the U cancel each other out, so the linear trend is flat even though complaints rise sharply when the pool is much colder or much warmer than 27 °C.

Task 4: The outlier alibi

r

```
ggplot(case, aes(lifeguards, sunscreen)) +
  geom_point(size = 2.5) +
  geom_point(data = case[13, ], colour = "red", size = 4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sunscreen sales vs lifeguards on duty",
       x = "Lifeguards", y = "Sunscreen sales (units)") +
  theme_minimal()

cor(case$lifeguards, case$sunscreen)
cor(case$lifeguards[-13], case$sunscreen[-13])
```

- **The wrong record is row 13** (lifeguards = 16, sunscreen = 179). The other days have between 6 and 11 lifeguards, so 16 is far outside the range, and 179 is the highest sales figure by a wide margin.
- **r with all rows = 0.60** (moderate).

- **r without row 13 = 0.11** (weak).

That single record inflated r from about 0.11 to 0.60. The apparent link between lifeguards and sunscreen sales comes almost entirely from that one point.

Task 5: The prime suspect

r

cor(case$icecream$, case$rescues$)

cor(case$temp$, case$icecream$)

cor(case$temp$, case$rescues$)

```
cor(resid(lm(icecream ~ temp, data = case)),
   resid(lm(rescues ~ temp, data = case)))
```

- r(icecream, rescues) = **0.82**
- r(temp, icecream) = **0.92**
- r(temp, rescues) = **0.87**
- **Partial correlation controlling for temp = 0.12**

Verdict: The link between ice-cream sales and rescues falls from 0.82 to about 0.12 once temperature is held constant, so the two are connected mainly because hot days drive both. Ice cream does not cause rescues; temperature is the hidden driver. A **randomised controlled experiment** could test causation, for example randomly assigning ice-cream availability across beaches on days of similar temperature and comparing rescue calls. An alternative is a matched or stratified observational design that compares days within narrow temperature bands.

get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]

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3. The deceptive zero. Plot complaints against pool_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot sunscreen against lifeguards. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. icecream and rescues are strongly linked. Report r(icecream, rescues), then r(temp, icecream) and r(temp, rescues). Compute the partial correlation of icecream and rescues controlling for temp: correlate resid(lm(icecream ~ temp)) with resid(lm(rescues ~ temp)). Write a two-sentence verdict: does icecream cause rescues? Name one study design that could test